

Math 421

Friday, November 7

Announcements

- Homework 8 Due **today**
- Exam 2 – 11/12 5:45-7:15 pm
 - **Date:** Wednesday, 11/12
 - **Time:** 5:45-7:15 pm
 - **Location:** Birge 145
- Exam Week Drop-In Hours
 - Tuesday 1:30-3:30 pm VV B205
 - Wednesday 12:00-1:00 pm VV B205

Discussion Questions/Activity

1. What would you need to do to prove that $\lim_{x \rightarrow a} f(x)$ does not exist?

$$\lim_{x \rightarrow a} f(x) \neq L$$

a) According to the $\epsilon - \delta$ definition of the limit
There exists $\epsilon_0 > 0$ such that for all $\delta > 0$ for some x , $0 < |x - a| < \delta$
 $|f(x) - L| \geq \epsilon_0$.

* b) According to the sequence definition of the limit
There exists $(x_n) \subseteq S$, $x_n \neq a$ for all n , $\lim x_n = a$ but
 $\lim f(x_n) \neq L$. There exists $(x_n) \& (y_n)$, $\lim x_n = \lim y_n = a$
but $\lim f(x_n) \neq \lim f(y_n)$

2. Show that $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ does not exist, using 1.b).

$$\begin{aligned} \bullet (f(x_n)) &= (0, 0, 0, \dots) \\ &= (\sin \pi, \sin 2\pi, \sin 3\pi, \dots) \end{aligned}$$

$$f(x_n) = \sin(n\pi) \quad x_n = \frac{1}{n\pi}$$

$$\begin{aligned} \bullet (f(y_n)) &= (1, 1, 1, \dots) \\ &= \left(\sin\left(\frac{\pi}{2} + 2\pi\right), \sin\left(\frac{\pi}{2} + 2(2\pi)\right), \right. \\ &\quad \left. \sin\left(\frac{\pi}{2} + 3(2\pi)\right), \dots\right) \end{aligned}$$

$$f(y_n) = \sin\left(\frac{\pi}{2} + n(2\pi)\right)$$

$$y_n = \frac{1}{\frac{\pi}{2} + n(2\pi)}$$

Algebraic Limit Laws

Theorem. Let f and g be functions defined on a domain S and let a be the limit of some sequence in S . If the limits $L = \lim_{x \rightarrow a} f(x)$ and $M = \lim_{x \rightarrow a} g(x)$ exist and are finite. Then

- i. $\lim_{x \rightarrow a} kf(x)$ exists and equals kL for all $k \in \mathbb{R}$.
- ii. $\lim_{x \rightarrow a} (f + g)(x)$ exists and equals $L + M$.
- iii. $\lim_{x \rightarrow a} (fg)(x)$ exists and equals LM .
- iv. $\lim_{x \rightarrow a} (f/g)(x)$ exists and equals L/M provided $M \neq 0$ and $g(x) \neq 0$ for $x \in \text{dom}(g)$.

• ε, δ - "small"
finite a, L

• M, N - "large"
 $M > 0, N > 0 \rightarrow +\infty$
 $M < 0, N < 0 \rightarrow -\infty$

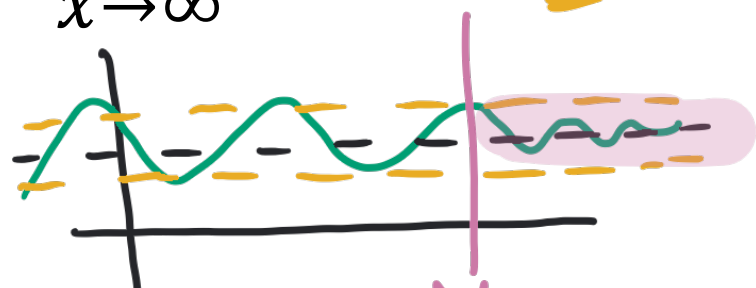
One-Sided & Infinite Limits

- $\lim_{x \rightarrow a^+} f(x) = \underline{L}$ for all $\varepsilon > 0$ there exists $\delta > 0$ such that $0 < x - a < \delta$ implies $|f(x) - L| < \varepsilon$.

$x > a$

$x \rightarrow a^-$
 $x < a$

- $\lim_{x \rightarrow \infty} f(x) = \underline{L}$ for all $\varepsilon > 0$ there exists $N > 0$ such that $x > N$ implies $|f(x) - L| < \varepsilon$.



- $\lim_{x \rightarrow a} f(x) = \infty$ for all $M > 0$ there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies $f(x) > M$



Activity

Consider each of the following limits, fill in the following blanks for each:

for each $\overbrace{M > 0}^{M < 0}$ there exists $\overbrace{\delta > 0}^{N < 0}$ such that
 $\overbrace{0 < a - x < \delta}^{x < N}$ implies $\overbrace{f(x) > M}^{f(x) < M}$

① $\lim_{x \rightarrow a^-} f(x) = \infty$

② $\lim_{x \rightarrow -\infty} f(x) = -\infty$

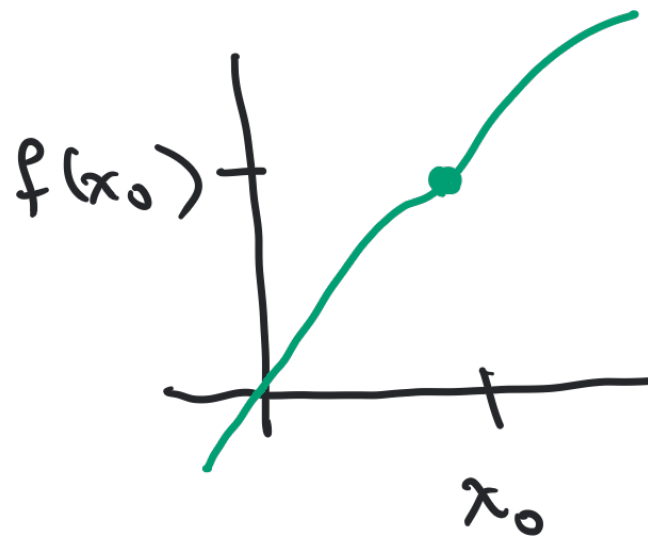
Chapter 3: Continuity

Section 17: Continuous Functions

Continuity

Definition. A function $f(x)$ is said to be **continuous at a point** $x_0 \in \text{dom } f$ if $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

If $f(x)$ is continuous at every $x_0 \in S$, then $f(x)$ is said to be a **continuous on S** . If $S = \text{dom } f$, $f(x)$ is called a **continuous function**.



Example

1. Recall we proved $\lim_{x \rightarrow 1} x^2 + x + 1 = 3$. Is $x^2 + x + 1$ continuous at 1?

yes!

2. Recall we proved $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$. Define function $g(x)$ so that $g(x) = x^2 \sin\left(\frac{1}{x}\right)$ for all $x \neq 0$ and $g(x)$ is continuous at all $x \in \mathbb{R}$.

$$g(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Activity – Discontinuity

Sketch at least three different ways a function can fail to be continuous at a point a .

